



The legal challenges for a Mars Base

Establishing a permanent human settlement on the Red Planet may be made complicated by international treaties on how space can be used <http://digit.in/marslegal>



How to build cities on Mars

Private space will be a concern in the early days, and as the base builds up, there will be a need for public spaces <http://digit.in/marscities>



HOW TO LIVE OFF THE LAND ON OTHER WORLDS

Similar to the first colonists visiting new lands, these missions will have to use the resources available locally to support their efforts

Aditya Madanapalle |
aditya@digit.in

Current space missions are relatively simple. Most of the resources needed for the mission are loaded on to the spacecraft before it leaves the Earth. At the moment, apart from what they take up from the Earth, spacecraft only use the energy from the Sun.

There are now incipient plans to undertake more ambitious missions. These include spaceships that hop between planets, pioneering manned missions to the outer solar system, and establishing permanent bases on the Moon and Mars.

Taking all the required resources from the Earth would be prohibitively

expensive. The pioneers and colonists will have to depend on locally available resources for at least some of their needs. The approach is known as In Situ Resource Utilisation or ISRU for short.

PATHFINDING MISSIONS

NASA plans to send a pioneering manned mission to Mars in the 2030s. NASA and SpaceX are both in the planning phase of a number of pathfinding missions to identify suitable landing spots.

SpaceX plans to use its Dragon spacecraft, dubbed the Red Dragon for the Mars pathfinding missions. The idea is to scout the planet and identify suitable landing spots. NASA's Mars Reconnaissance Orbiter (MRO) is also scanning the surface of the planet for signs of resources. Once promising sites are identified, NASA plans to dispatch rovers to the

locations, to confirm the presence of the required resources.

The site has to be near the equator to allow for optimum utilisation of the solar energy reaching the solar panels on the surface. At the same time, there is a requirement for water ice, preferably closer to the surface. The problem is that it is difficult to find large deposits of water ice in the equatorial regions, where the solar energy received is at its maximum.

NASA's Lunar Crater Observation and Sensing Satellite (LCROSS) was an impact lander that slammed into the surface of the Moon in 2009 and found a surprising amount of water near the south pole of the Moon. NASA is planning the Resource Prospector mission, to validate technologies for harvesting of hydrogen, oxygen and water from the Moon. If successful, it would be the first interplanetary mission to demonstrate mining of local resources.

ROCKET FUEL

There are three kinds of rocket propellants that can be manufactured by ISRU. Liquid oxygen and liquid hydrogen have been used in the Space Shuttle and the Saturn V rocket that took humans to the moon. The Japanese H-IIA and the European Ariane 5 launch vehicles both use liquid oxygen and liquid hydrogen. Hydrogen peroxide is used for altitude control on spacecraft. Engines that use liquid oxygen and methane are in development.

On the Moon, it is possible to electrolyse the water ice to produce liquid oxygen and liquid hydrogen. The challenges involved are identifying the sites where abundant water ice is found close to the surface, and extracting it. On both the Moon and Mars, the water ice can be used to produce hydrogen peroxide, which can be used to power not only the rockets, but also the rovers and industrial equipment such as the drills necessary to extract the resources.

SpaceX plans to build a fuel depot on Mars that will use the Sabatier